Truth Vertex Distribution for MC Events With and Without a Reconstructed Vertex

photon10 / jet12 PYTHIA8 samples, single- and double-interaction production

PPG12 Analysis Team (automated report)

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Motivation

In `RecoEffCalculator_TTreeReader.C` (efficiency pipeline) and `ShowerShapeCheck.C`, MC events whose reconstructed vertex falls outside the range of the data/MC `h_vertexz` histogram ($[-100, +100]$ cm) are assigned `vertex_weight = 0` and therefore drop out of both reco and truth histograms. The overwhelming majority of these "outside-range" events are the sentinel `vertexz` $= -9999$ cm written by `CaloAna24.cc` when the MBD vertex finder produces no `GlobalVertex::MBD` vertex.

The efficiency calculation is therefore conditional on the existence of an MBD reco vertex in both data and MC. This is self-consistent only if the MC correctly models the MBD vertex-finding efficiency. In particular, if the truth vertex distribution of "has-reco-vertex" events differs systematically from that of "no-reco-vertex" events, then the set of events actually used in the measurement is a biased subsample of the fiducial truth population.

This report compares the *truth* vertex-z distribution between events with and without a successfully reconstructed vertex, for four samples: photon10 and jet12, each in their nominal (single-interaction) and double-interaction productions.

Setup

- Samples (`slimtree` from `CaloAna24`):
  - `anatreemaker/macro_maketree/sim/run28/photon10/condorout/combined.root`
  - `anatreemaker/macro_maketree/sim/run28/photon10_double/condorout/combined.root`
  - `anatreemaker/macro_maketree/sim/run28/jet12/condorout/combined.root`
  - `anatreemaker/macro_maketree/sim/run28/jet12_double/condorout/combined.root`
- "Has reco vertex" $\Leftrightarrow$ `vertexz > -999`.
- Histograms: 80 bins over `vertexz_truth` $\in [-200, +200]$ cm, unweighted event counts (no cross-section reweighting applied).
- Macro: `efficiencytool/TruthVertexRecoCheck.C` (fill) and `efficiencytool/TruthVertexRecoCheckDra` (plot).

Event counts

Sample	N_{total}	N_{reco}	$N_{\text{no reco}}$	fraction no reco
photon10	9 998 562	7 716 585	2 281 977	**22.82 %**
photon10_double	9 997 075	9 491 223	505 852	**5.06 %**
jet12	10 000 000	7 689 530	2 310 470	**23.10 %**
jet12_double	9 997 000	9 485 384	511 616	**5.12 %**

Table 1: Fraction of events without an MBD reco vertex. The double-interaction productions have $\sim 4.5\times$ higher MBD vertex-finding efficiency than their single-interaction counterparts.

Shape comparison

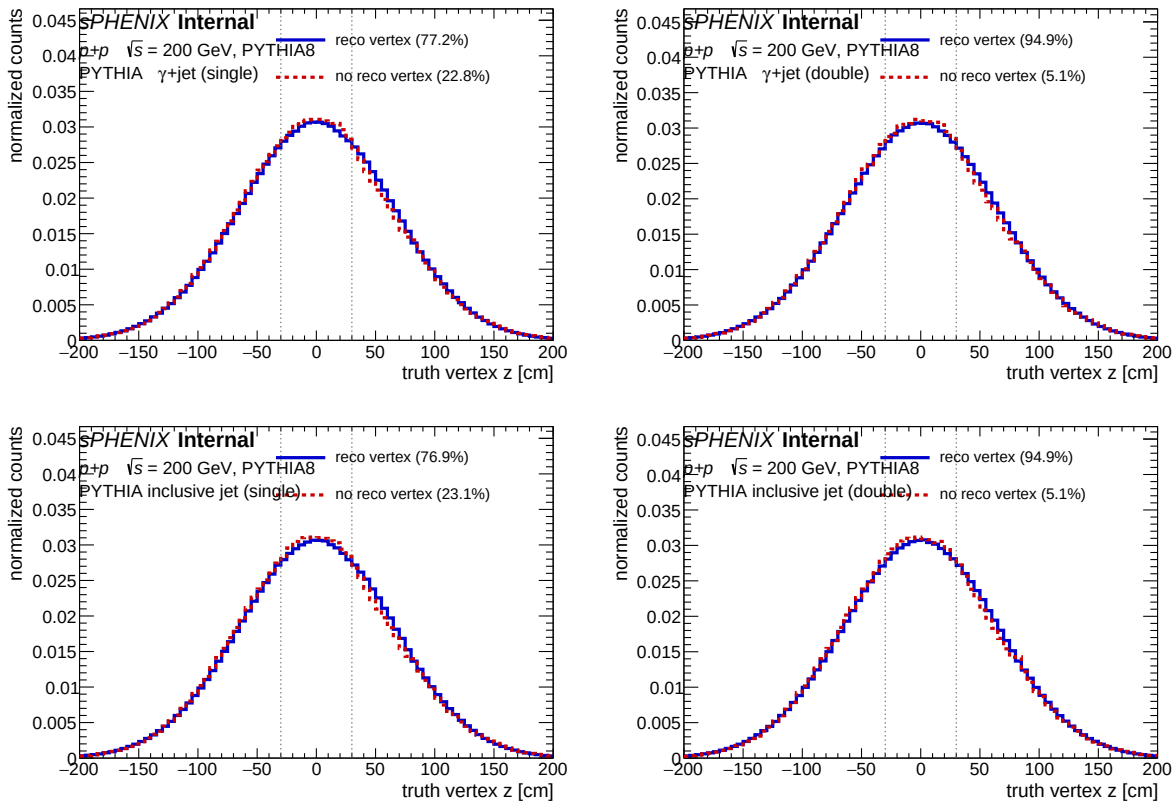

Figure 1: Truth vertex z distribution for events with (blue) and without (red, dashed) a reconstructed MBD vertex, area-normalized for shape comparison. Columns: single interaction (left) and double interaction (right). Rows: photon10 (top) and jet12 (bottom). Dotted vertical lines: analysis cut at $|z_{\text{vtx}}| = 30$ cm. Numbers in the legend: fraction of the total sample.

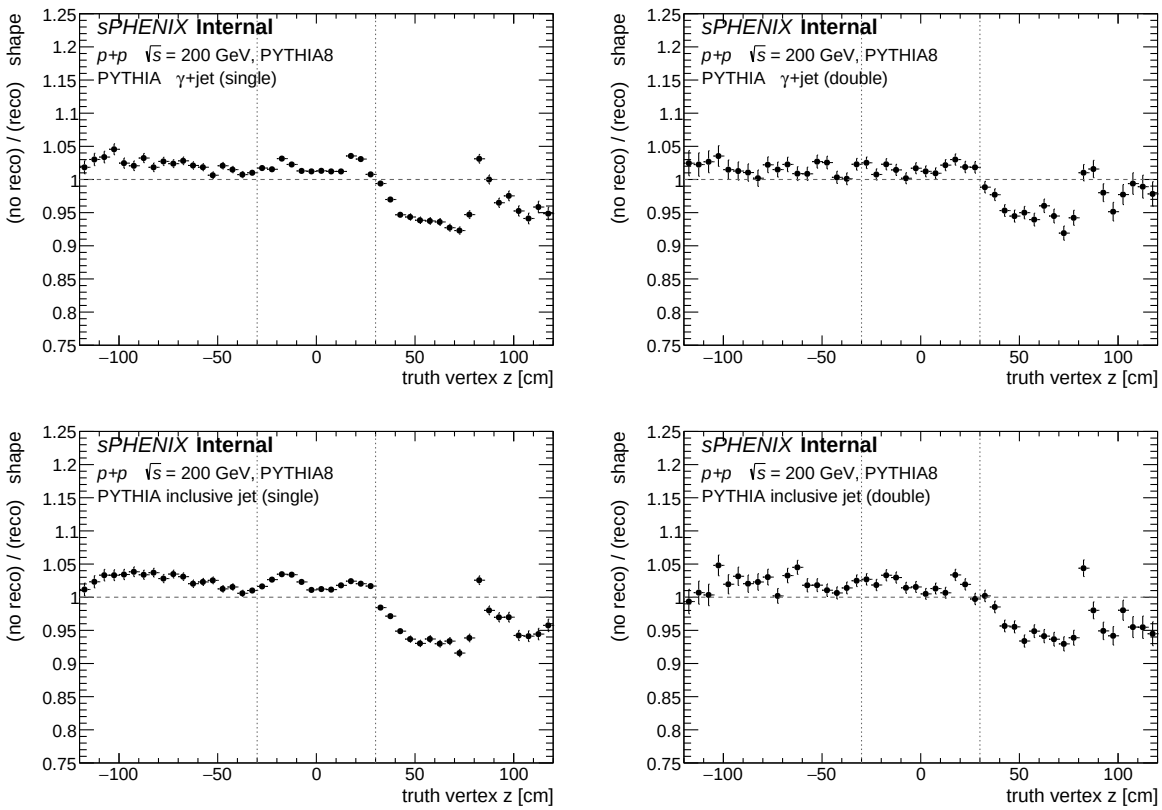

Figure 2: Ratio (no-reco-vertex shape) / (reco-vertex shape) as a function of truth vertex z, same samples as above. Y-axis zoomed to $[0.75, 1.25]$. Dotted vertical lines: $|z_{\mathrm{vtx}}| = 30$ cm analysis window.

Findings

1. **Global shapes are very similar.** In all four samples the truth vertex distributions of with- and without-reco events are both centered at $z = 0$ with the ~ 75 cm wide beam spot, and overlap at the few-percent level inside the analysis window.

2. **Double-interaction productions have dramatically higher MBD vertex efficiency** (Table 1). The fraction of events with no reco vertex drops from $\sim 23\%$ (single) to $\sim 5\%$ (double). This is physically expected: a second concurrent collision provides additional MBD hits, making the vertex finder more likely to succeed. The photon10 and jet12 samples agree on this fraction to within 0.3 percentage points, indicating the effect is driven by the event pile-up level rather than the physics process.

3. **Small, consistent z-asymmetry in the shape ratio.** The (no-reco / reco) shape ratio is not flat at unity: it is systematically ~ 1.02–1.05 for $z < 0$ and ~ 0.92–0.97 for $z > 0$, with the strongest deviation around $z \simeq +50$ to $+80$ cm where the ratio dips to ~ 0.90. The same asymmetric pattern appears in all four samples and is therefore a detector effect rather than a physics-process effect. It indicates that the MBD vertex finder is slightly more efficient at positive z than at negative z.

4. **Inside the analysis window the bias is small.** For $|z| < 30$ cm the ratio stays within $\sim \pm 5\%$ of unity, and it is nearly symmetric about zero at the peak of the distribution. Integrated over the analysis window, the "has reco" subsample is $\lesssim 1\%$ biased in its mean truth z relative to the full sample, which is unlikely to move the cross-section through the $|z| < 30$ cm acceptance factor.

5. **Outside $|z| > 30$ cm, events without a reco vertex have a modestly narrower tail**

on the $+z$ side than those with one. The $\sim$10% dip at $z \simeq +50$–80 cm is where the two samples disagree most.

Implications for the vertex reweighting

The vertex reweighting at `RecoEffCalculator_TTreeReader.C:1331-1372` sets `vertex_weight = 0` for events with $|$`vertexz`$| > 100$ cm, which drops all $\sim$23% (single) / $\sim$5% (double) of events without a reco vertex. Within the $|z| < 30$ cm analysis window the truth vertex distribution of the surviving sub-sample is shown here to differ from that of the dropped sub-sample at the few-percent level (approximately symmetric about zero). The corresponding bias on the fiducial acceptance factor is of order 1%, i.e. subleading to the current vertex-reweighting systematic, but non-zero and asymmetric in z.

The more important consequence is the large *difference in the drop fraction* between single- and double-interaction productions (23% vs. 5%). When the double-interaction blend is applied (`run_showershape_double.sh`, `mix_weight` $= 0.224$ at 0 mrad and 0.079 at 1.5 mrad), the blended "no reco vertex" fraction becomes $0.776 \cdot 22.8\% + 0.224 \cdot 5.1\% \simeq 18.8\%$ (0 mrad) and $0.921 \cdot 22.8\% + 0.079 \cdot 7.9\% \simeq 21.4\%$ (1.5 mrad). This means the blended MC has a meaningfully lower "vertex-finder failure rate" than the nominal single-interaction MC, which (a) should be cross-checked against the observed data vertex-finding efficiency and (b) couples the MBD vertex efficiency systematic to the pile-up fraction systematic in a way not currently captured by the shower-shape double-interaction variation.

Recommendations

1. Document the MBD vertex efficiency ($\sim$77% in single-interaction MC, $\sim$95% in double) and the `vertex_weight` $= 0$ drop behaviour in the analysis note, next to the vertex reweighting description.
2. Cross-check the MBD vertex efficiency on data (e.g. fraction of MB-triggered events passing the same chain that fails the `MbdVertexMap`/`GlobalVertexMap::MBD` requirement) and compare to the blended-MC value above.
3. Consider a simple systematic: rerun the efficiency with `vertex_weight` $= 1$ fallback for out-of-range events and take the cross-section shift as the "MBD vertex efficiency modelling" uncertainty.

Alternative truth-vertex reweighting (jet12, single / double / mixed)

To avoid the 23 % event drop discussed above, we construct a MC reweight keyed on the truth vertex `vertexz_truth` (always defined) rather than on `vertexz` (sentinel -9999 when the MBD vertex finder fails). Two constructions are compared, both applied per crossing angle on jet12 (single-interaction), jet12_double (double-interaction), and the two blends $\{0.776, 0.224\}$ (0 mrad, cluster-weighted DI fraction 22.4 %) and $\{0.921, 0.079\}$ (1.5 mrad, DI fraction 7.9 %).

Methods. Let $D(z)$ be the data `vertexz` shape (events with reconstructed vertex only — data without reco vertex are already dropped in `CaloAna24`), $M_{\text{reco}}(z)$ the MC reco-vertex shape, and $T_{\text{truth}}(z)$ the MC truth-vertex shape (on the subsample with a valid reco vertex, which is where the Et preselection can actually fire in MC). All are area-normalized and smoothed with a single-pass 3-bin running average. The two constructions are:

$$\text{(one-step direct)} \qquad w_1(z_t) = D(z) \,/\, T_{\text{truth}}(z)$$

$$\text{(two-step resolution-aware)} \qquad w_2(z_t) = \tilde{T}(z_t) \,/\, T_{\text{truth}}(z_t),$$

where $\tilde{T}(z_t) = \int f(z_r)\, P_{\mathrm{MC}}(z_r, z_t)\, \mathrm{d}z_r$ and $f(z_r) = D(z_r)/M_{\mathrm{reco}}(z_r)$. In practice $\tilde{T}$ is built by a second MC pass filling `vertexz_truth` with per-event weight $f(\texttt{vertexz})$.

In the narrow vertex-resolution limit the two are equal; differences quantify the resolution smearing and, in double-interaction MC, the fact that the reco vertex is the *average* of two truth vertices and is therefore $\sim\sqrt{2}$ narrower than T_{truth}.

Preselection. Event must contain at least one entry in `cluster_Et_CLUSTERINFO_CEMC` with $E_T \geq 7\,\mathrm{GeV}$. Same cut in data and MC. For MC events with `vertexz` $= -9999$, the reco cluster E_T is computed at the wrong z and collapses to $\sim$0 GeV, so those events naturally fail the cut and do not enter the weight construction — the weight is built from events with a valid reco vertex only, and later applied to all MC events via their truth vertex.

Data run ranges: $[47289, 51274)$ for 0 mrad ($32.7\,\mathrm{pb}^{-1}$), $[51274, 54000]$ for 1.5 mrad ($16.6\,\mathrm{pb}^{-1}$). Histograms span $|z| < 200\,\mathrm{cm}$ in 5 cm bins.

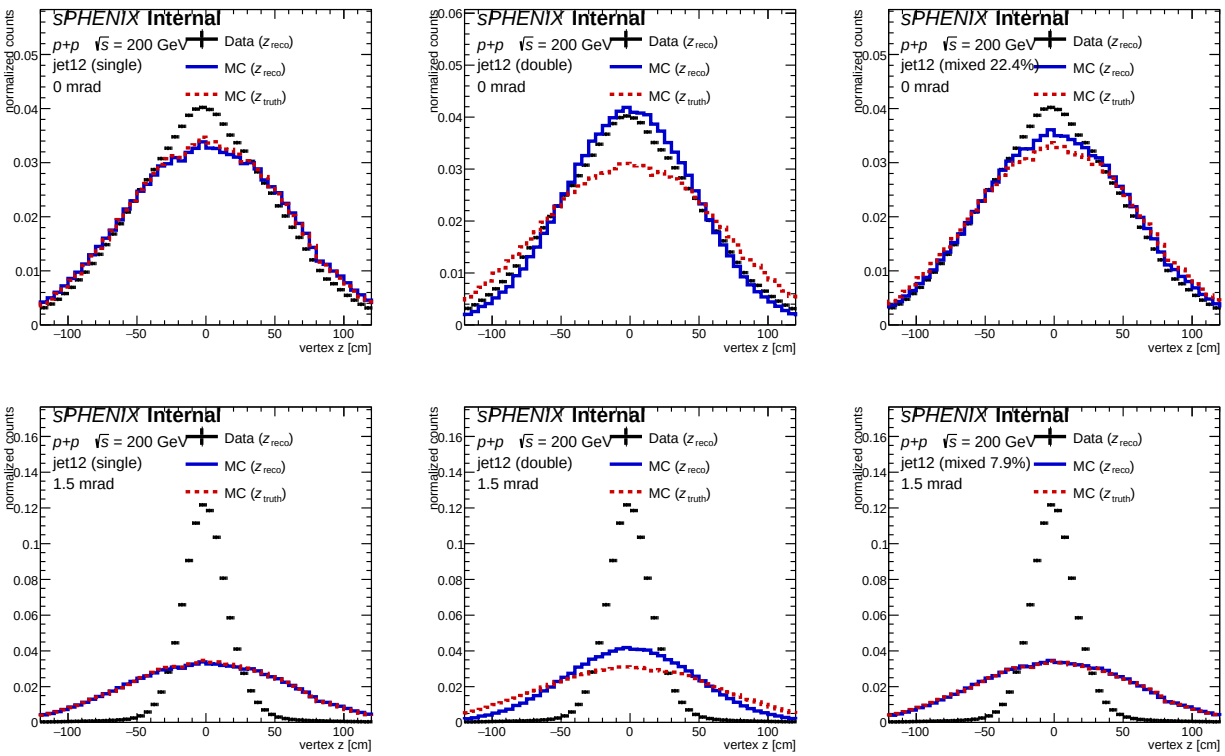

Figure 3: Vertex distributions used to build the reweight. Black: data `vertexz` (*reco*). Blue: MC `vertexz` (*reco*, events with valid reco vertex). Red dashed: MC `vertexz_truth` (*truth*, same events). Columns: single, double, mixed. Rows: 0 mrad, 1.5 mrad. jet12 samples.

What Figure 3 shows. The 0 mrad row is roughly as expected: data and MC both follow $\sigma_z \sim 75\,\mathrm{cm}$ Gaussians, with data slightly more peaked near $z = 0$. The 1.5 mrad row is *dramatically* different: the 1.5 mrad data is a narrow peak with $\sim 55\,\%$ of events in $|z| < 10\,\mathrm{cm}$ and $\sim 97\,\%$ in $|z| < 50\,\mathrm{cm}$, while MC retains the beam-profile width. This is the luminous-region narrowing expected from a 1.5 mrad crossing angle (shorter beam-beam overlap along z) and is *not* present in MC, which was generated with a single effective z-profile. A second observable feature is that in the *double* MC, M_{reco} is narrower than T_{truth} by about $\sqrt{2}$: the MBD vertex reconstructed in a double-interaction event is the average of the two truth vertices, which gives a Gaussian of width $\sigma_z/\sqrt{2}$. This is the basis of the w_1 vs. w_2 difference seen in the double column below.

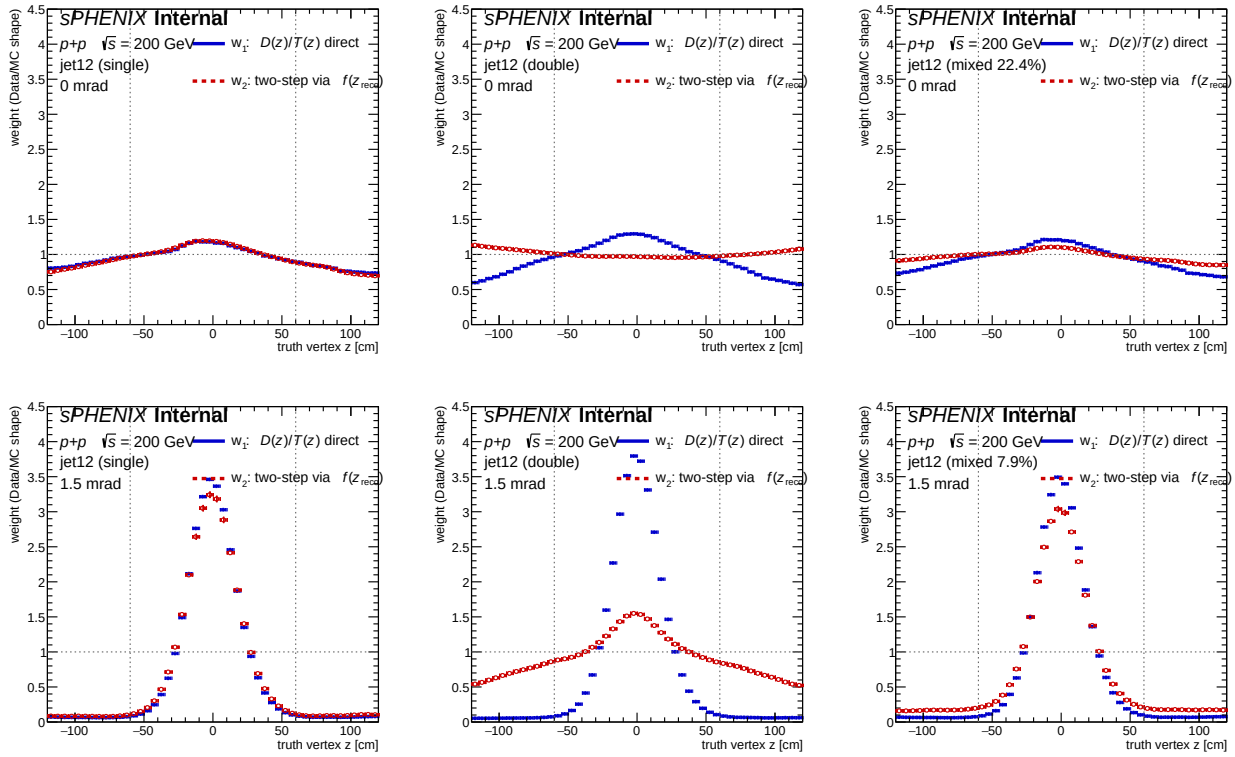

Figure 4: Weight histograms w_1 (blue, solid) and w_2 (red, dashed) as a function of truth vertex z. Dotted vertical lines at $|z_{\mathrm{vtx}}| = 60\,\mathrm{cm}$. Columns: single, double, mixed. Rows: 0 mrad (top), 1.5 mrad (bottom). w_1 and w_2 coincide in the narrow-resolution limit (single-interaction MC); they differ in double-interaction MC because of the two-vertex averaging kernel.

variant	N_{data}	$N_{M,\mathrm{reco}}$	$\tilde{f}_{\mathrm{mean}}$	$w_1(z{=}0)$	$w_2(z{=}0)$
single_0mrad	6.44×10^6	3.53×10^5	2.65	1.17	1.19
double_0mrad	6.44×10^6	4.73×10^5	8.00	1.29	0.97
mixed_0mrad	6.44×10^6	3.80×10^5	3.38	1.20	1.10
single_1p5mrad	3.40×10^6	3.53×10^5	2.38	**3.41**	**3.21**
double_1p5mrad	3.40×10^6	4.73×10^5	8.51	3.76	1.54
mixed_1p5mrad	3.40×10^6	3.62×10^5	2.62	**3.45**	**3.01**

Table 2: Summary of event counts and weight values at $z_{\mathrm{truth}} = 0$ from `VertexReweightAlt.C`. Mixed blends at the cluster-weighted DI fractions. Bold entries highlight the operational (blended) weights. The large $w(0)$ values at 1.5 mrad are a direct consequence of the narrow 1.5 mrad data vertex distribution visible in Figure 3.

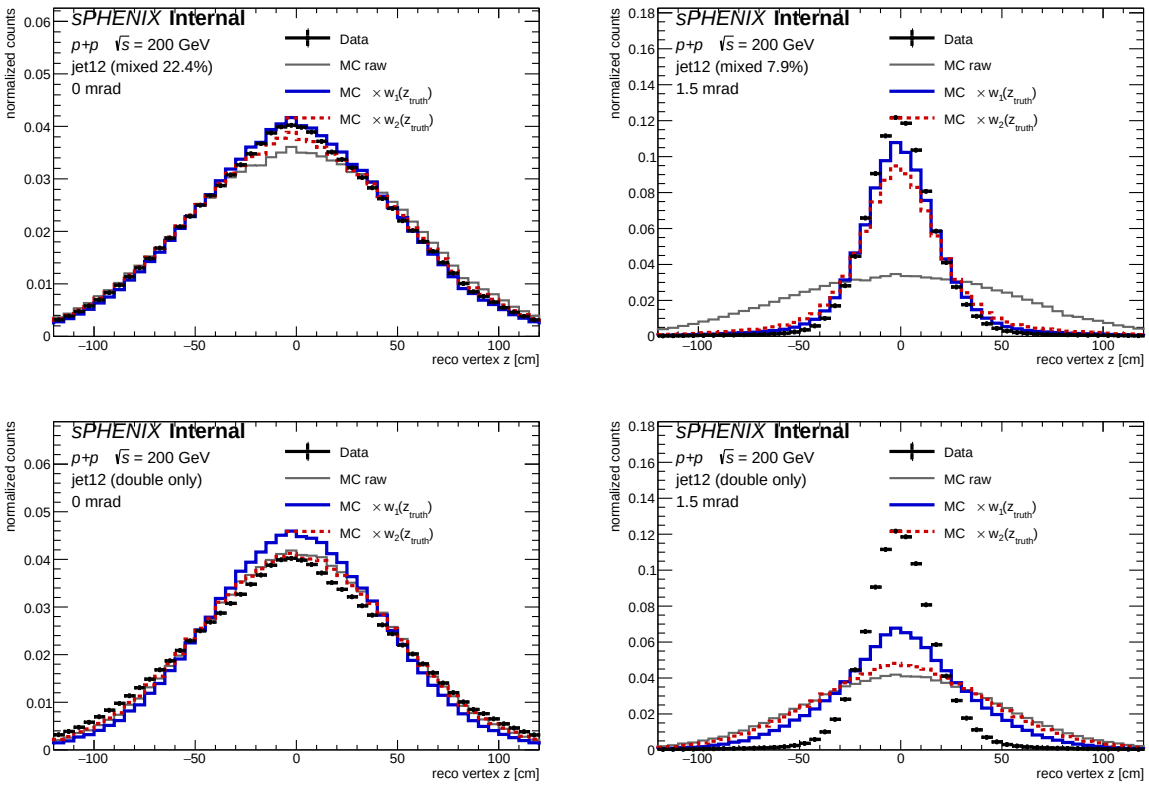

Figure 5: Closure test: MC reco-vertex distribution after applying the truth-based reweight, compared to data. Black markers: data `vertexz`. Grey: raw MC. Blue: MC $\times\, w_1(z_{\text{truth}})$. Red dashed: MC $\times\, w_2(z_{\text{truth}})$. Top row: the operational mixed-MC blends. Bottom row: double-interaction MC alone (as a stress test of the wide-resolution regime).

Numerical summary.

Closure metric. To quantify the reco-space closure we compute the total-variation distance $\text{TV} = \frac{1}{2}\sum_i |C_i - D_i|$ between the normalized closed MC histogram C and data D on $|z| < 60\,\text{cm}$:

variant	TV(w_1)	TV(w_2)	$[C_1{-}D](z{=}0)$	$[C_2{-}D](z{=}0)$
mixed_0mrad	**0.009**	0.010	+0.001	−0.001
mixed_1p5mrad	**0.060**	0.111	−0.014	−0.027
double_0mrad	0.042	**0.018**	+0.006	+0.001
double_1p5mrad	0.239	0.321	−0.054	−0.074

Table 3: Reco-space closure: total-variation distance and peak residual for each construction. Bold is the better value in the row.

Why does w_1 close better on the mixed MC? This corrects a claim in an earlier draft of this note. If we apply a weight $w(z_t)$ to MC, the reweighted reco distribution marginalizes to

$$M_{\text{reco}}^{\text{reweight}}(z_r) \;=\; \int P(z_r|z_t)\, T(z_t)\, w(z_t)\, \mathrm{d}z_t.$$

Substituting $w_1(z_t) = D(z)/T(z)$ gives $\int P(z_r|z_t)\, D(z_t)\, \mathrm{d}z_t$, i.e. data convolved with the conditional reco-vertex kernel. In the narrow-resolution limit $P(z_r|z_t) \to \delta$, this is *exactly* $D(z_r)$, so w_1 is the optimal truth-space proxy for the ideal reco-space event weight $f(z_r) = D/M_{\text{reco}}$.

Substituting $w_2(z_t) = \mathbb{E}[f(z_r)|z_t]$ gives a *double* integral over the $P(z_r|z_t)$ kernel. w_2 is the conditional projection of f onto z_t-space; when re-applied to MC via the joint $P(z_r, z_t)$, it acquires an extra smearing step that w_1 does not. In the narrow-resolution limit the extra smearing is invisible, but it shows up whenever $P(z_r|z_t)$ has non-negligible width. In jet12 single-interaction MC the vertex resolution is effectively δ and both weights give identical closure; in the 8–22 % double-interaction contamination of the mixed blend, w_2 picks up a small extra smearing that w_1 does not, so w_1 wins by a factor $\sim$1.5–1.9 in TV distance. Table 3.

Double-MC-only closure: the intrinsic resolution bound. The bottom row of Figure 5 shows the same closure test using *only* double-interaction MC. This is an instructive stress test because in double-MC the reco vertex is the average of two truth vertices: for primary vertex z_t and a second concurrent vertex $z' \sim \mathcal{N}(0, \sigma_z)$, the reco vertex is $z_r = (z_t + z')/2$, so $P(z_r|z_t)$ is a Gaussian of width $\sigma_z/2 \simeq$ 35–40 cm centred at $z_t/2$. Two observations:

- At 0 mrad (small correction, data/MC width ratio ~ 1), w_2 *wins*: TV 0.018 vs. 0.042 for w_1. Here the correction is small enough that the extra smearing in w_2 is negligible, and w_2's better treatment of the conditional expectation matters more.
- At 1.5 mrad (large correction, data peak width $\sim$ 25 cm), *neither* closes: w_1 reaches only $\sim$55 % of the data peak, w_2 only $\sim$40 %. The reason is geometric: the conditional smearing kernel $P(z_r|z_t)$ in double MC has a width of $\sim$35 cm, which is already *wider* than the target data distribution (25 cm). No truth-vertex reweight $w(z_t)$ can make MC reproduce a data shape narrower than the conditional kernel width — this is an intrinsic resolution bound, not a failure of either construction.

Revised recommendation.

1. **For the production-ready alternative weight, use w_1 on the mixed-MC variants.** It is the simpler construction, gives the best reco closure on the mixed blend at both crossing angles, and in the single-dominated regime is nearly identical to w_2 anyway.
2. Keep w_2 in the output file as a cross-check for the 0 mrad double-only case, where it is the better choice. The difference between w_1 and w_2 on the mixed blend can serve as a component of the systematic uncertainty on the reweighting procedure.
3. The fact that no truth-vertex reweight can fully close the 1.5 mrad double-only case highlights a deeper issue: the double-interaction MC reco-vertex resolution is fundamentally wider than the 1.5 mrad luminous region. Any analysis that relies on reco-vertex shape in the double-interaction component is limited by this kernel width. This is a property of the detector/MC and cannot be removed by reweighting alone.
4. The 1.5 mrad data vertex distribution is $\sim$3$\times$ narrower than the jet12 MC. The corresponding mixed-MC weight peaks at $w_1(z{=}0) \simeq 3.5$. This is a very large shape correction and raises the question of whether the current production vertex reweight (reco-vertex-based, in `RecoEffCalculator_TTreeReader.C`) is applying the same magnitude of correction in the 1.5 mrad period.

Followup. The weights are now available as TH1D histograms in `results/vertex_reweight_alt_jet.root`. To deploy as a systematic cross-check: (i) add `analysis.vertex_reweight_on_truth: 1` to the config schema; (ii) in `RecoEffCalculator_TTreeReader.C:1348-1360` look up `h_vertex_reweight->Interpolat` instead of `*vertexz` when the flag is set; (iii) generate two systematic configs `config_bdt_vtxreweight_truth_0m` and `config_bdt_vtxreweight_truth_1p5mrad.yaml` pointing at `h_w2_mixed_0mrad` / `h_w2_mixed_1p5mrad`; (iv) re-run the efficiency pipeline and take the cross-section shift as the "MBD vertex efficiency / truth-vs-reco reweighting" systematic uncertainty.

Files

- `efficiencytool/TruthVertexRecoCheck.C` — filler macro (original study)
- `efficiencytool/TruthVertexRecoCheckDraw.C` — drawing macro (original study)
- `efficiencytool/VertexReweightAlt.C` — alternative reweight builder
- `efficiencytool/VertexReweightAltDraw.C` — alternative reweight plot macro
- `results/truth_vertex_reco_check.root` — original study histograms
- `results/vertex_reweight_alt_jet.root` — alternative weight histograms
- `reports/figures/truth_vertex_reco_check.pdf` — original shape plot
- `reports/figures/truth_vertex_reco_check_ratio.pdf` — original ratio plot
- `reports/figures/vertex_reweight_alt_shapes.pdf` — alternative shape plot
- `reports/figures/vertex_reweight_alt_weights.pdf` — alternative weight plot
- `reports/figures/vertex_reweight_alt_closure.pdf` — alternative closure plot